Aim

To implement hashing using a hash function and resolve collisions using linear probing technique in C program.

Algorithm – Insert

1. Calculate index = key % TABLE_SIZE.

2. If the calculated position is empty, insert the key.

3. Otherwise, move to the next position.

4. Repeat until an empty position is found.

5. Use modulo TABLE_SIZE to wrap around.

Algorithm – Search

1. Calculate index = key % TABLE_SIZE.

2. Check the calculated position.

3. If the key is found, return success.

4. If the position is empty, key does not exist.

5. Otherwise move to the next position.

6. Continue until the starting position is reached.

Program

```
#include <stdio.h>
#include<stdlib.h>
#define TABLE_SIZE 10
int h[TABLE_SIZE]={NULL};
void insert()
{
 int key,index,i,flag=0,hkey;
 printf("\nenter a value to insert into hash table\n");
 scanf("%d",&key);
 hkey=key%TABLE_SIZE;
 for(i=0;i<TABLE_SIZE;i++)     {
    index=(hkey+i)%TABLE_SIZE;
    if(h[index] == NULL)      {
       h[index]=key;
        break;
    }
   }
```

```
   if(i == TABLE_SIZE)
    printf("\nelement cannot be inserted\n");
}
void search()
{
 int key,index,i,flag=0,hkey;
 printf("\nenter search element\n");
 scanf("%d",&key);
 hkey=key%TABLE_SIZE;
 for(i=0;i<TABLE_SIZE; i++)
 {
   index=(hkey+i)%TABLE_SIZE;
   if(h[index]==key)
   {
     printf("value is found at index %d",index);
     break;
   }
 }
 if(i == TABLE_SIZE)
   printf("\n value is not found\n");
}
void display() {

 int i;
 printf("\nelements in the hash table are \n");
 for(i=0;i< TABLE_SIZE; i++)
 printf("\nat index %d \t value =  %d",i,h[i]);
}
int main() {
   int opt,i;
   while(1)    {
      printf("\nPress 1. Insert\t 2. Display \t3. Search \t4.Exit \n");
      scanf("%d",&opt);
      switch(opt)       {
         case 1:
            insert();
            break;
         case 2:
            display();
            break;
         case 3:
            search();
            break;
         case 4:exit(0);
      }
   }
}
```

OUTPUT:

```
Press 1. Insert  2. Display     3. Search   4.Exit
1
enter a value to insert into hash table
34

Press 1. Insert  2. Display     3. Search       4.Exit
1

enter a value to insert into hash table
45

Press 1. Insert  2. Display     3. Search       4.Exit
2

elements in the hash table are

at index 0      value =  0
at index 1      value =  0
at index 2      value =  0
at index 3      value =  0
at index 4      value =  34
at index 5      value =  45
at index 6      value =  0
at index 7      value =  0
at index 8      value =  0
at index 9      value =  0
Press 1. Insert  2. Display     3. Search       4.Exit
3
enter search element
45
value is found at index 5
Press 1. Insert  2. Display     3. Search       4.Exit
4
```

Result:

Thus program has been implemented in C for hashing using a hash function and linear probing technique and operations are performed successfully.